

Information Transmitting Efficiency in China Gold Spot Market: a Random Walk Test with Wild and Bootstrap Variance Ratio Method

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Abstract--China provides an excellent empirical case study on the information transmitting efficiency in the capital market. Our paper extend the existing investigation based on much more new data and test the efficiency of information transmitting with Individual, Sub-sampling and Wild bootstrap Variance Ratio method. The empirical results show that gold prices reflect all the past information efficiently and no one can get abnormal return continuously. The China gold spot price followed Random Walk and the gold spot market is a weak-form efficiency market.

Keywords: *Information Efficiency; Variance Ratio; Wald Bootstrap; EMH*

1. INTRODUCTION

As well as other emerging markets, China provides an excellent empirical case study on the information transmitting efficiency in the capital market. In the efficiently market, assets prices follow Random Walk and always reflect all available relevant information instantaneously and fully, where the information sets consist of all past prices and returns. As a result, assets prices are purely unpredictable based on past information and fluctuate only in response to the random flow of new information (Fama, 1970)[1]. Much more literatures of Efficiency market Hypothesis (EMH) had concerned on the test of Random Walk in China stock market. However, there is no research which specifically investigates the question: does the gold spot market are efficiently or China gold price follow a random walk? This paper will extend the existing investigation in two ways: First, our study will be based on much more new data of the gold market; Second, the efficiency of information transmitting is evaluated with multi-Variance Ratio method issued by Kim(2006)[2], which are more powerful than others especially when the data are small samples. The results showed that the entire gold prices follow Random Walk and had reflected all the past information at least during the empirical time. The statistic improved that no one can get abnormal return continuously. Obviously, the China gold spot market is weak-form efficiency.

The remainder of the paper is laid out as follows. In the next section we provide the short review about the Variance Ratio method and the third section will issue the methodology about Variance ratio. In the forth section will report the data, the fifth section will discussed the results and the conclusion follows in the end.

2. LITERATURE RIVIEW

Reviews of the EMH empirical study show that the Variance Ratio methodology (Lo and Mackinlay, 1988)[3] is a class test for RWH¹. The Variance Ratio methodology consists of testing the random walk hypothesis against stationary alternatives by exploiting the fact that the variance of random walk increments is linear in all sampling intervals, i.e., the sample variance of q-period return is k times the sample variance of one period return. Lo and MacKinlay had verified that Variance ratio test is more reliable than traditional random walk test. They suggested using overlapping data in computing the variance of long-horizon returns because it can potentially improve the test power. After Lo and MacKinlay, Chow and Denning (1993)[4] suggested multiple Variance Ratio tests to avoid sequential procedure leads to size distortions problem, but this method was also critical to note that the asymptotic and limiting sample will result bias and right skewness. Wright(2000)[5], in order to circumvent this problem, introduced alternative Variance Ratio tests based on ranks and signs, his nonparametric variance ratio are better than others. Whang and Kim (2003)[6] and Kim (2006) proposed two alternatives which do not rely on asymptotic approximations by suggesting sub-sampling and a wild bootstrap version of the Chow-Denning test. Al-khazali(2007)[7] test the RWH in Emerging Market with multi-Variance Ratio method, Kim(2008)[8] reviewed most of the research of Variance Ratio and test the Asian stock market, Kim (2009)[9] re-modified Auto Variance Ratio test and test it again. Thomas and B. M. Lucey(2008)[10] test the market efficiency of MENA region with Multi-Variance Ratio. Their results all showed Variance Ratio method is very appropriate for emerging market to test

¹ See Amâdie Charles et al.(2009), Variance-Ratio Test of Radom Walk: An Overview, Journal of Economic Surveys, Vol.23, No.3, 503-527; Hoque et al.(2007), A comparison of variance ratio tests of random walk: A case of Asian emerging stock markets. International Review of Economics and Finance, 16, 488-502.

the information efficiency and Random walk Hypothesis in the capital market. To the best of our knowledge, Graham Simth(2002) firstly used three prices with multiple Variance Ratio test empirical the random walk Hypothesis for London gold price[11]. However, Variance Ratio was not used to test the Efficiency Market Hypothesis in China gold spot market, even other empirical results with different method in future market are mixed conclusions. This support us strongly to take Random Walk test for China gold market and discuss the information transmitting efficiency deeply.

3. METHODLOGY

A. Random Walk Hypothesis

If a time series of returns follows a random walk then in a finite sample the increments in the variance are linear in the observation interval. We assume the gold market is efficient; all effects on gold price variations due to changes of other asset prices will be contemporaneous. Information on past price changes of other financial assets is, therefore, not useful in predicting. The classic definition of "Random walk process" to describe the movement of the capital prices is

$$P_t = \mu + P_{t-1} + \varepsilon_t, \varepsilon_t \sim IID(0, \sigma^2) \quad (1)$$

Where P_t is the gold price at time t , μ is the change or the arbitrary drift item of the price. The $IID(0, \sigma^2)$ means that ε_t is independent identically distributed, its mean is zero and the variance is σ^2 . The testing of the Random Walk Hypothesis would be divided into two steps. Step one is to test the RWH with Variance Ratio by Lo and MacKinlay, Step two is to test the RWH with Kim's test and Wright's test during the different period.

B. Individual Variance Ratio Test

Variance Ratio test is created by Lo and MacKinlay which is widely used to test the hypothesis of random walk. The most advantage of this method is that it permits the series not to be normal distribution necessarily but also to be heteroskedasticity either. Let P_t is the gold price, there will be

$$Var(P_t - P_{t-q}) = q[Var(P_t - P_{t-1})] \quad (2)$$

Here Var is the variance, q is any positive integer. Function (3) gives the definition of Variance Ratio:

$$VR(q) = \frac{1}{q} \frac{Var(P_t - P_{t-q})}{Var(P_t - P_{t-1})} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho(k) \quad (3)$$

Lo and Mackinlay(1988) and Liu and He(1991) gave a modified function of new Variance Ratio:

$$VR(q) = \frac{1}{q} \frac{\sigma^2(q)}{\sigma^2(1)} = \frac{\frac{1}{m} \sum_{t=1}^{nq} (P_t - P_{t-q} - \hat{\mu})^2}{\frac{1}{nq-1} \sum_{t=1}^{nq} (P_t - P_{t-1} - \hat{\mu})^2} \quad (4)$$

The gradual variance of $VR(q)$ is equal to $\Phi(q)$

$$\Phi(q) = \frac{2(2q-1)(q-1)}{3q(nq)} \quad (5)$$

And the normal distribute statistic value $Z(q)$ is

$$Z(q) = \frac{VR(q) - 1}{[\Phi(q)]^{1/2}} \sim N(0,1) \quad (6)$$

When the ε_t is heteroskedasticity the $VR(q)$ is still close to one in probability. Then the gradual variance is

$$\Phi^*(q) = \sum_{j=1}^{q-1} \left[\frac{2(q-j)}{q} \right]^2 \hat{\delta}(j) \quad (7)$$

Here

$$\hat{\delta}(j) = \frac{\sum_{t=j+1}^{nq} (P_t - P_{t-1} - \mu)^2 (P_t - P_{t-q} - \hat{\mu})^2}{\left[\sum_{t=1}^{nq} (P_t - P_{t-1} - \hat{\mu})^2 \right]^2}$$

So the $Z^*(q)$ is

$$Z^*(q) = \frac{VR(q) - 1}{[\Phi^*(q)]^{1/2}} \square N(0,1) \quad (8)$$

Here $\hat{\mu}$ = Average return, we will use both Homoscedastic and heteroscedastic test statistics for aggregation values q of 2, 4, 8, and 16.

C. Wild Bootstrapping Test by Kim

Kim(2006) proposed a wild bootstrap approach to improve the small sample properties of Variance Ratio test. His approach involves computing the Lo-Mackinlay and Chow-Denning test statistics on samples of T observations formed by weighting the original data by mean zero and variance one random variables and using the bootstrap distribution the test statistics. The main advantage of this finite sample test does not rely on asymptotic approximations. Therefore, it is robust to nonnormality. The test statistic can be written as:

$$MV(q_i) = \max |z^*(q_i)| \quad (9)$$

The wild bootstrap variance ratio test can be conducted in three stages as followed:

Step I: Form a bootstrap sample of T observations $\alpha_t^* = \eta_t \alpha_t, (t=1, 2, \dots, T)$, where η_t is a random sequence with zero mean and unit variance; a normal distribution is used here;

Step II: Calculate $MV(q_i)$ using α_t^* from the bootstrap sample generated in Step I;

Step III: Repeat step I and step II m times, for example (unconfirmed) time in this paper, to form a bootstrap distribution of the best statistic $MV(q_i, j)_{j=1}^m$.

The bootstrap distribution $MV(q_i, j)_{j=1}^m$ is used to approximate the sampling distribution of $z^*(q)$ given in equation (8). In the Monte Carlo simulations, Kim demonstrated that wild bootstrapping variance ratio tests are powerful and robust alternatives for testing the random-walk hypothesis.

4. DATA DESCRIPTION

Our data are obtained from Shanghai Gold Exchange. Gold prices are expressed as currency unit per RMB. The open and close price of Gold Au99.99 and Gold Au99.95 are examined. Empirical period is begging from 30th Oct 2002 to 31st Dec 2010.

Table:1 Statistics of return of Gold Price

Series	Gold95 Open Price	Gold95 Close Price	GOLD99 Open Price	GOLD99_C Close Price
Mean	0.000643	0.000645	0.000648	0.000642
Min	-0.087573	-0.062010	-0.093786	-0.085331
Std.	0.012396	0.010921	0.012839	0.011678
Skew	-0.011865	0.539268	0.153414	0.293697
Kurt	12.36486	15.31315	16.54388	17.39491
J-Bera	15057.17	17080.32	7191.488	12565.91
Pro	0.000000	0.000000	0.000000	0.000000
Q(12)	35.03	11.356**	37.020	17.375*
Q(24)	44.558	33.384*	54.967	40.567
Ljung-Box Q of Squared Return				
Q(12)	98.741	168.04	50.468	49.790
Q(24)	169.59	216.89	217.40	216.56

*t, test the mean equal to zero; Kurtosis, coefficient of excess kurtosis. Q(q) is the Ljung-Box Q statistic for q-order serial correlation, which distributes as a chi-square variate with q degrees of freedom. *, ** Statistically significant at the 10% and 5% levels, respectively.

In Table-1 we summarized the statistics of the gold returns calculated as the first differences in the logarithms of the gold prices and Figure-1 show the daily volatility of GoldAu99.99 and GoldAu99.95 prices. The result showed closely characters, Gold99 and Gold95 (hereinafter) have close standard deviation and all the means are close to zero. The data are all leptokurtic; all the

returns are not normal distribution. Furthermore, the excess kurtosis in all cases are much higher than zero (as required by normal distribution), which indicates that the empirical distributions of the returns have fat tails.

According to Chow-Denning(1993), statistical inference for randomness using the variance ratio test should be based on the heteroskedasticity-adjusted statistic. Table-1 also provides the Ljung–Box Q statistics for testing serial correlations. The results show that all the returns series are significantly serially correlated. We also computed the Ljung–Box Q statistic for the squared returns for checking heteroskedasticity. As can be seen in Table 1, the Ljung–Box Q statistics for the squared returns are substantially larger than those for the raw returns, indicating the existence of heteroskedasticity.

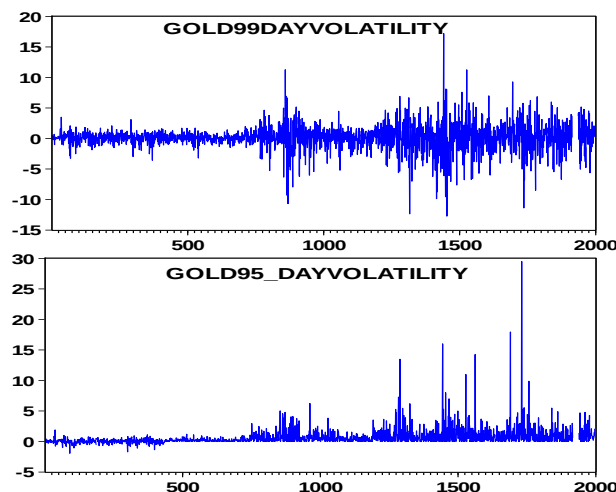


Figure-1 Day volatility of GoldAu99.95

5. EMPIRICAL RESULT AND ANALYSIS

All the results of Lo and MaKinlay show in Table-2, z-Statistics shows the statistics in case of homoscedasticity and z*-Statistics is in case of heteroscedasticity. Highlight letters indicate the rejected results of $z(q)$ statistics or $z^*(q)$ statistics. For the entire gold prices, we use unbiased variances, heteroskedasticity assumptions and allowing the series are *IID*; secondly, we use homoscedasticity assumptions version of the Lo-MacKinlay test with no bias correction. All tests chose the aggregation level from 2 to 16. Under the heteroskedasticity assumption, Gold99_close price rejected the null hypothesis for aggregation levels of $q=2$, other samples accept the null hypothesis; but under the homoscedasticity assumption, Gold99_open price, Gold95_close and Logarithm Gold95_close price accept the null hypothesis, Logarithm Gold99_Open and Gold99_close price rejected the null hypothesis for aggregation levels of $q=2$, Logarithm Gold99_lose price rejected the null hypothesis for aggregation level of $q=4$ as well as Gold95_open price and Logarithm Gold95_open price do.

At the same time, the $z(q)$ values of Logarithm Gold99_open price and Logarithm Gold95_open price are higher with the increasing of the lag period after $q=2$, but it is smaller with the increasing of the lag period in Logarithm Gold99_close price and Logarithm Gold95_close. It means the information efficiency is changing between different gold. Others series have the irregular values. Whatever the assumption is, the Gold99_close price rejected the null hypothesis, the price does not follow the random walk.

Table 3 summaries the p-values for the wild bootstrap test by Kim for Gold99 and Gold95 price, we also use the standard error estimates assume, no heteroskedasticity assuming and computing variances assuming zero mean, chose biased variance estimates and lags from 2 to 16. As we reviewed ahead, Kim's bootstrap test has more power than all the others test. All the p-Value is smaller than the Lo and MacKinlay's test. Gold99_close price rejected the null hypothesis again; the others price accept the random walk hypothesis. This means the information transmitting in China Gold Spot Market is efficiency.

Table-2 Lo and MacKinlay Test Result

Gold99_Open Price			Logarithm Gold99_Open Price		
	z-Statistic	z*-Statistic		z-Statistic	z*-Statistic
2	-1.000120	-0.628809	2	-3.177071	-1.602575
4	-1.647101	-1.087375	4	-2.501988	-1.416279
8	-1.800948	-1.198277	8	-1.996080	-1.252114
16	-1.588721	-0.998038	16	-1.511384	-0.951787
Gold99_Close Price			Logarithm Gold99_Close Price		
	z-Statistic	z*-Statistic	Period	z-Statistic	z*-Statistic
2	3.526771	2.440542**	2	2.207734	1.515890
4	2.723205	1.892825	4	2.330473*	1.610543
8	1.180212	0.886536	8	1.432385	1.069404
16	0.510864	0.492259	16	1.034594	0.870921
Gold95_Open Price			Logarithm Gold95_Open Price		
	z-Statistic	z*-Statistic		z-Statistic	z*-Statistic
2	-2.729427	-1.116688	2	-3.517223	-1.570841
4	-3.827490	-1.797935	4	-3.763172	-1.905517
8	-2.965009	-1.594626	8	-2.515395	-1.428366
16	-2.416081	-1.398700	16	-1.886183	-1.127471
Gold95_Close Price			Logarithm Gold95_Close Price		
	z-Statistic	z*-Statistic		z-Statistic	z*-Statistic
2	-1.486398	-0.440958	2	-0.692410	-0.316662
4	-1.155532	-0.385154	4	-0.113269	-0.002853
8	-1.377770	-0.543543	8	-0.145417	0.008637
16	-1.333252	-0.566917	16	-0.152617	0.050233

*** Statistically significant at the 5% and 10% level, respectively;
z-Statistic is Homoscedasticity; z*-Statistic is Heteroscedasticity

Table-4 summaries the daily results for the two sub-period sample. We used Whang(2003)¹ test to take the number of block lengths from an equally spaced grid in the interval($2.5T^{0.3}$, $3.5T^{0.6}$) for the sub-sampling test, at the same time we list the results of Kim test to compare the results with heteroskedasticity standard error estimates assume, the number of bootstrap replications m is set to1000, the aggregation levels are the same as Chow-Denning test. All the sub-samples do not reject the null hypothesis. So it means the gold price follow random walk in all the sub-periods. This is a strong support for our hypothesis; because we can't give the inference which the market being efficiency if the sub-periods are not efficiency while the whole period is negative.

Table3- Variance Ratio Tests using Bootstrapping

Lag	Gold99 open	Gold99 Close	Log Gold99 open	Log Gold99 Close
2	0.5450	0.0110	0.1100	0.1220
4	0.3210	0.0540	0.1640	0.1040
8	0.2600	0.3120	0.2360	0.2370
16	0.3750	0.6180	0.3970	0.3190
	Gold99 open	Gold99 Close	Log-Gold99 open	Log Gold99 Close
2	0.2620	0.6870	0.1230	0.7830
4	0.0680	0.7230	0.0570	0.9640
8	0.1400	0.5890	0.1670	0.9280
16	0.2030	0.5890	0.2950	0.8480

Standard error estimates assume no heteroskedasticity; Compute variances assuming zero mean Use biased variance estimates ; User-specified lags: 2 4 8 16;The number of bootstrap replications m is set to1000 Test probabilities computed using wild bootstrap: dist=twopoint,

¹ Whang and Kim(2003) develop a multiple Variance Ratio test which uses a subsampling technique of Politis et al.(1997), which is a data-intensive method of approximating the sampling distribution.

Table-4 Variance Ratio Tests using Sub-Sampling

Sub-period I	Glod95 open	Glod95 close	Glod99 open	Glod99 close
Chow-Denning MV	0.86493	1.439737	1.29122	1.590542
Kim(2006) P-value	0.7780	0.4340	0.3810	0.2140
Block Lengths				
49	0.8440	0.3460	0.8450	0.7190
77	0.4200	0.2680	0.5040	0.5350
105	0.6180	0.3660	0.7110	0.7650
133	0.5530	0.3820	0.5980	0.6480
161	0.4810	0.3600	0.4390	0.4950
Sub-period II				
Chow-Denning MV	1.574406	1.234051	0.795439	0.899701
Kim(2006) P-values	0.2560	0.4140	0.7260	0.6360
Block Lengths				
50	0.1800	0.2220	0.4240	0.3740
77	0.2820	0.3020	0.5000	0.4940
104	0.3090	0.3370	0.4540	0.4810
131	0.4110	0.4160	0.5450	0.5640
158	0.5150	0.5140	0.6280	0.6100

The 5% and 10% critical values for Chow-MV statistic are 2.49 and 2.23, respectively; Following Whang and Kim (2003), block lengths are chosen from an equally spaced grid in the interval $[2.5T^{-1/2}, 3.5T^{-1/2}]$. Sub-period I covers the period from 30/10/2002 to 08/11/2005, 744 data; Sub-period II covers from 09/01/2008 to 31/12/2010, 727 data.

6. CONCLUSION

With a brief review of Variance Ratio test of the random walk hypothesis, our paper examined the information transmitting and weak-form efficiency for the China gold spot market with Gold Au99.99 and Gold Au99.95. We used individual Variance Ratio test by Lo and MacKinlay, Sub-period test by Wright and Wild Bootstrap Method by Kim. The results are more clearly than other mixed research that only Gold99_close price rejected the null hypothesis strongly. At the heteroskedasticity assumption, most prices accept the null hypothesis by Lo and MacKinlay test. But at the homoscedasticity assuming we didn't agree with this result that most results reject the null hypothesis because it is a strictly assumption for the price series. Generally, we still consider that the China gold market is weak-form efficiency with our sample not only during the whole sample period but also the sub-periods. In generally, the information transmitting will be efficiently in the efficiently market.

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